

Statistics Unit 1: Introduction

Lecture Notes

Lobotomy

From 1948 to 1952, about 5,000 per year in US

Neuropsychiatrist Walter Freeman and Neurosurgeon James Watts

Egas Moniz [e'gush mô'nēsh], developed Leucotomy, won Nobel prize

Research in 1935:

- 7 completely recovered
- 7 improved somewhat
- 6 unchanged

3 additional cases published in 1937. . .

Case I: B.V.J., female, aged 36, depressed, history of suicide attempts, paranoid delusions

After Leucotomy: On the 12th day she felt well: "It's all over. I've been locked up now for five and a half years but I wasn't crazy. I want to go back to live with my daughters." Was in good condition although possibly a little reticent. The family found her in excellent condition, just as she had been before the psychosis. She left the hospital and now after six months she remains in entirely normal condition.

Case II: J.R.G, aged 36, male, police officer, agitated, paranoid delusions, catatonic behavior, tearing his clothes with his teeth.

After Leucotomy: After two weeks, "the patient was in good condition, talked relevantly and coherently, and was correctly oriented. He desired to go home in order to resume his work. He was observed by Prof. Cid who considered him well and permitted him to leave. Information obtained recently indicates that he is maintaining his recovery now seven months after operation.

Case III: J.J.S., aged 36, male, newsboy, excessive drinking, paranoid delusions, domestic violence, hearing voices, unsociable, odd behavior.

After Leucotomy: After three weeks, "it was noted that the patient was getting along well and we were convinced that the hallucinations of hearing had disappeared. We wished to send him to the hospital to be examined by Prof. Cid but the patient considered himself cured and he was sent home. After three months, the patient was altogether well and he had already undertaken his usual occupation.

What went wrong?

Need to Quantify Observations

Options

Categorical

dichotomous

not dichotomous

Numerically scaled

ordinal

interval

ratio

What if Moniz administered the Brief Psychiatric Rating Scale (BPRS)?

Scores range from 0 to 108, Clinical mean = 24 (sd = 11)

	BPRS score
Case 1, B.V.J.	27
Case 2, J.R.G.	36
Case 3, J.J.S.	54

What questions could we address?

1. Are these scores high or low?
2. Are these scores reliable?

Reliability: The same amount of something always produces the same assessed numerical value regardless of when or how it is assessed.

What is the correlation between Moniz and Prof. Cid?

	Moniz BPRS score	Prof. Cid BPRS score
Case 1, B.V.J.	27	36
Case 2, J.R.G.	36	33
Case 3, J.J.S.	54	60

3. Are these scores valid?

Validity: Scale measures what it is supposed to measure

If the BPRS is valid, with what should it correlate?

	Moniz BPRS score	Days worked in year
Case 1, B.V.J.	27	125
Case 2, J.R.G.	36	10
Case 3, J.J.S.	54	20

4. Correlational research design

- Definition: Study where scores for all the variables are determined by assessment and there are NO manipulated variables.
- Observe that which occurs naturally. No manipulation of variables
- Includes difference score; includes research process and etiology
- Limitation: determining cause

5. Experimental research questions

- Definition: Study that has at least one variable with scores determined by an assessment, and at least one manipulated variable produced by the researcher forcing cases into groups.
- Uses experimental manipulation and random assignment
- Includes clinical trial
- Limitation: explaining natural phenomena

What if Moniz conducted a clinical trial?

Leucotomy	BPRS score
Case 1, B.V.J.	27
Case 2, J.R.G.	36
Case 3, J.J.S.	54

$$\bar{x} = 39$$

Control	BPRS score
Case 4, C.W.K.	45
Case 5, K.S.H.	39
Case 6, K.K.T.	51

$$\bar{x} = 45$$

Difference: $39 - 45 = -6$

What questions could we address?

6. How big is the effect? Effect Size (descriptive)

7. Could it be a random chance? Inferential

Terms

Population

Parameter: Numerical value that is based on all the cases in an entire population.

Sample

Statistic: Numerical value that is based on cases from a sample.

Inferential statistic

Descriptive statistic

The mean

A group of something: Each group member has a score on a variable “x”

The group has a mean

$$\bar{X} = \frac{\sum X}{n}$$

A single case “i” The case has a score on a variable “x”

If x_i is unknown. . .

What is the best prediction of x_i ? What will make the least error?

The linear equation:

$$\hat{x}_i = \bar{x}$$

This entire course will be about this equation!

Note: Later, we will show why this is a linear equation.

The mean

If you pick someone at random and you want to bet
The mean is the score you expect to get

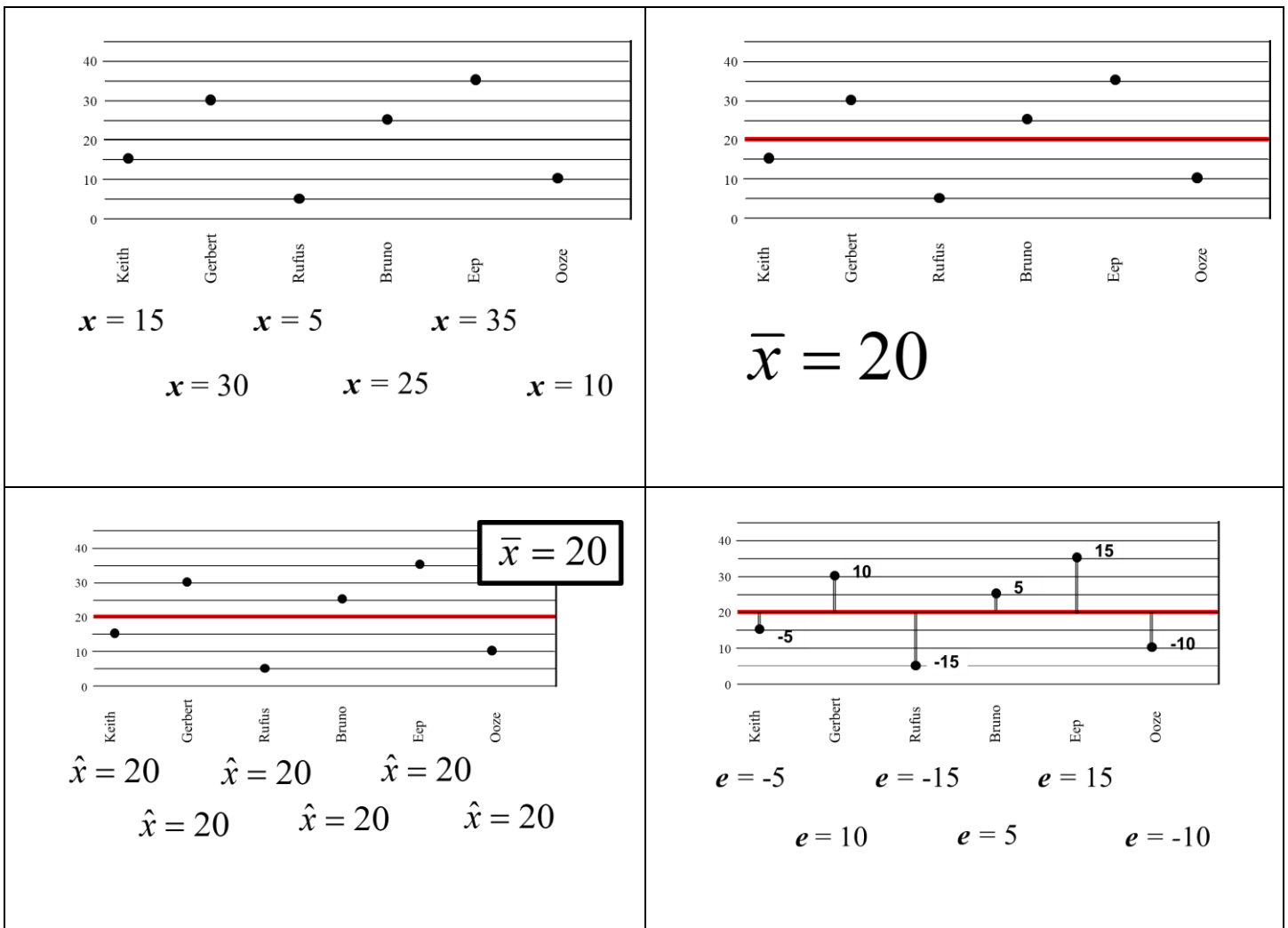
Linear equation without a hat

$$x_i = \bar{x} + e_i$$

x_i = The score on x for case i

$\bar{x}$ = The mean of all x_i

e_i = the error for case i
= how much case i deviates from the mean



$$x_i = \bar{x} + e_i$$

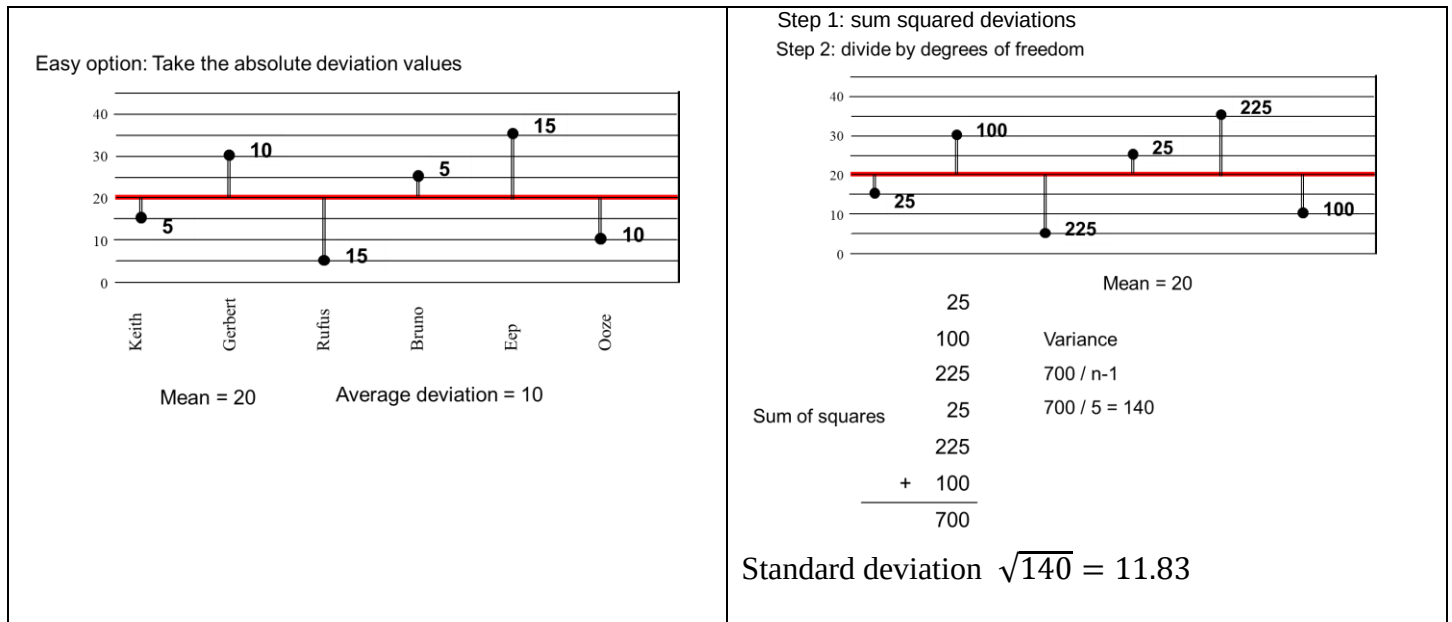
Every case has a score on x_i

$\bar{x}$ is the mean of x_i

Every case has an e_i

What is the average deviation of e_i ? or $|e_i|$? What would it tell you?

How would you calculate it?



Definition for degrees of freedom (what does her score have to be?)

Bias: “variance / n” underestimates parameter

Definitions:

Expected value: The average of all the values in a population of values from which a single value can be randomly sampled.

Bias: When the expected value for a statistic is not equal to the population parameter.

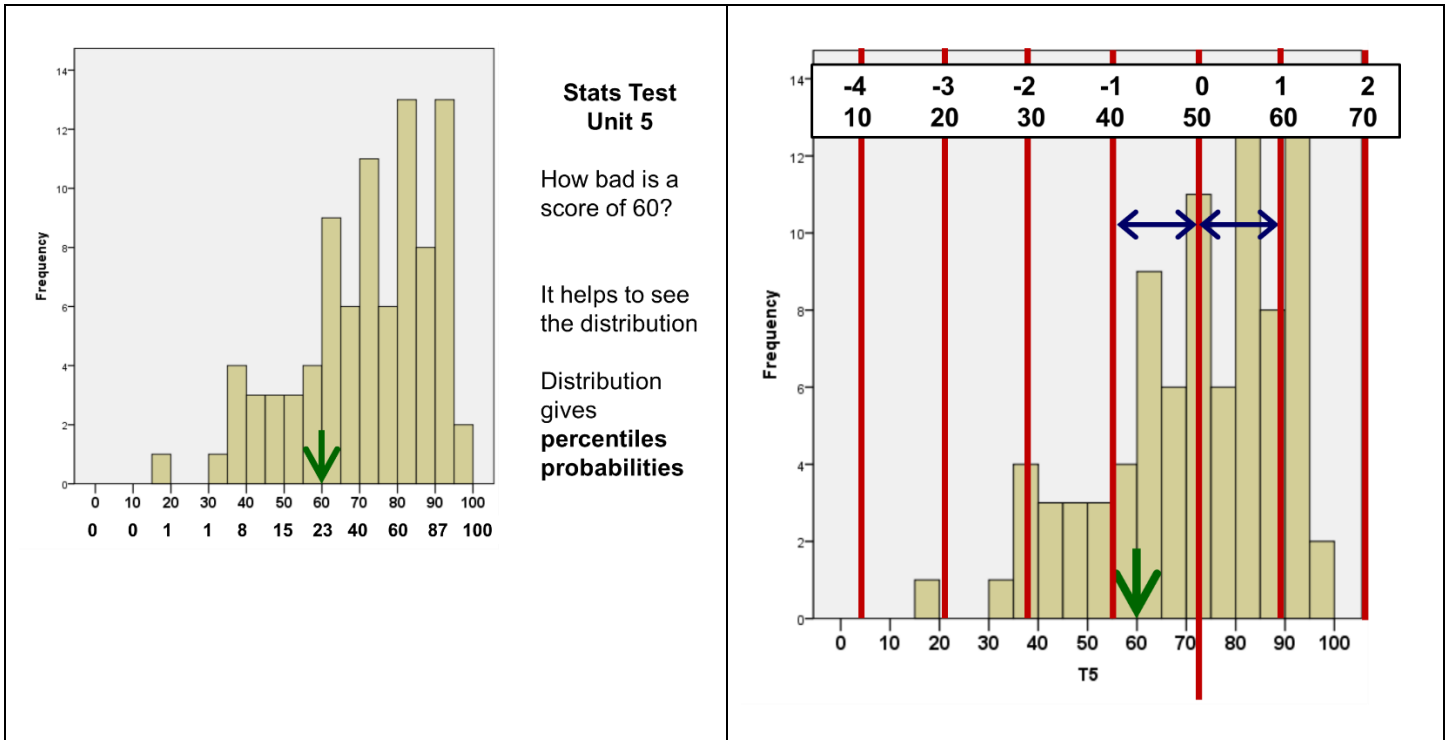
Sum of squares $SS = \sum (x_i - \bar{x})^2$

Variance $sd^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

Standard deviation $sd = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

The standard deviation

The standard deviation is used to measure things
It's how far the average person strays from the mean



Distributions described by mean and sd (What do you guess these are?)

Mean = 72 sd = 17

Scale can be measured with *sd* units

60 is .71 sd below mean

Scores in sd units are:

$$z = \frac{x_i - \bar{x}}{sd}$$

z-scores

Mean = 0

sd = 1

$$T = 10z + 50$$

T-scores

Mean = 50

sd = 10

Score = 60

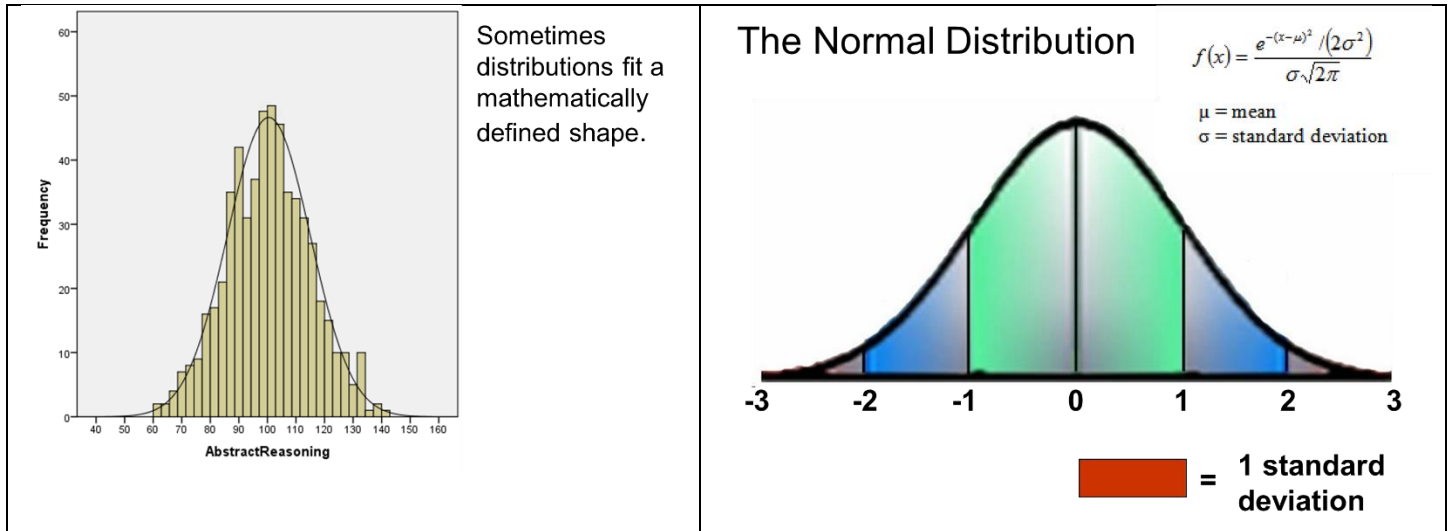
z-score = -.71

T-score = 42.9

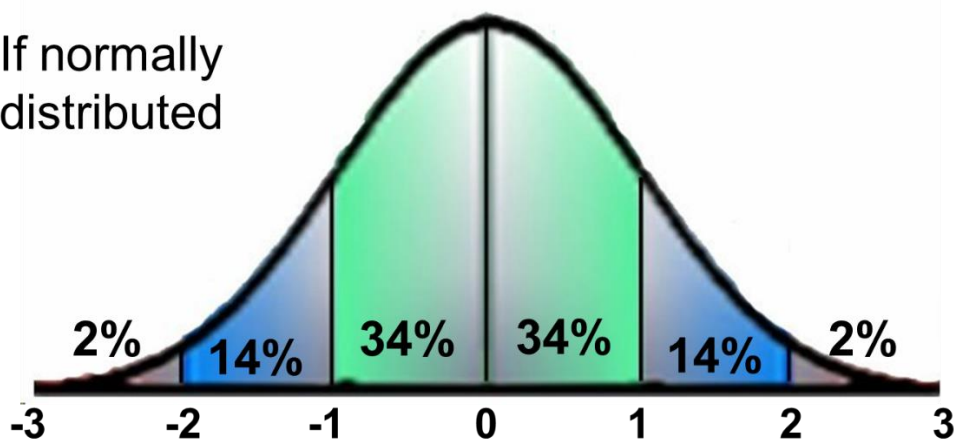
Terms:

centered score

standardized score



If normally distributed



- Percentile for score
- Probability of getting scores lower or higher
- Probability of getting scores more extreme

Lecture Add-On Notes

How happy are romantic couples in college?

307 college students in romantic relationships

Satisfaction
scale

We have fun together.
We provide comfort to each other.
We accept each other.
We help each other.
We resolve our conflicts.
We admire each other.
We understand each other.
We share personal thoughts and feelings.
We cooperate together.

1	2	3	4	5
Much less than I want	Less than I want	A little less than I want	Almost as much as I want	As much as I want

Written summary of mean and sd:

On a satisfaction scale that ranged from 1 to 5, the average score was 4.20. On average, people deviated from this score by .65 units.

Written summary of z-score:

Electra's satisfaction score was .74 standard deviations below the mean. If satisfaction is normally distributed, her score is between the 16th and 50th percentile.

Notes:

1. Do not use abbreviated SPSS labels in written summaries.
2. Round to two decimal places.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
satisfaction	307	1	5	4.20	.65
Valid N (listwise)	307				

